

Owner override and open-ended tutor contract

LEARNER CONDITIONING

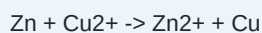
Student knowledge percentage was not supplied. This review fixture uses an explicit owner override; no learner percentage is fabricated. The override changes support only, not Chemistry truth or frozen question identity.

CORE2B JOB

Coach interpretation and move selection before releasing solution detail. The learner must commit to what is asked, relevant evidence, representation, invariant, first move and justification.

RQ-REDOX-01 | Commit before help

GOVERNED REACTION



FROZEN QUESTION

For $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$, prove which species is oxidised and which is reduced using oxidation-state changes.

YOUR PLAN

1 What is being asked? 2 Which species/state evidence matters? 3 Which representation will you use? 4 What must remain true? 5 What is your first move? 6 Why is that move legal?

RQ-REDOX-01 | Progressive concept help

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

H1 ORIENT

Name the target in your own words before calculating or labelling anything.

H2 STRUCTURE / H3 REPRESENT

Track same element -> state lane -> electron consequence. Keep role blank until self-change is secure.

H4 PRINCIPLE / H5 FIRST MOVE

Increase = oxidation; decrease = reduction. First move: write the two before/after lanes.

RQ-REDOX-01 | Solution, falsify, generalise

FULL SOLUTION

Zn: 0 -> +2, so Zn is oxidised. Cu in Cu^{2+} : +2 -> 0, so Cu^{2+} is reduced.

FALSIFY

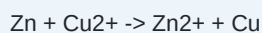
Try to disprove your answer: does each role agree with the species' own state/electron change? Does electron loss equal gain?

GENERALISE

State a method that would still work if Zn/Cu were replaced by unfamiliar species: identity -> state direction -> electron consequence -> role.

RQ-REDOX-02 | Commit before help

GOVERNED REACTION



FROZEN QUESTION

Write the two electron statements and identify the reducing and oxidising agents.

YOUR PLAN

1 What is being asked? 2 Which species/state evidence matters? 3 Which representation will you use? 4 What must remain true? 5 What is your first move? 6 Why is that move legal?

RQ-REDOX-02 | Progressive concept help

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

H1 ORIENT

Name the target in your own words before calculating or labelling anything.

H2 STRUCTURE / H3 REPRESENT

Track same element -> state lane -> electron consequence. Keep role blank until self-change is secure.

H4 PRINCIPLE / H5 FIRST MOVE

Increase = oxidation; decrease = reduction. First move: write the two before/after lanes.

RQ-REDOX-02 | Solution, falsify, generalise

FULL SOLUTION

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$. $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. Zn is the reducing agent; Cu^{2+} is the oxidising agent.

FALSIFY

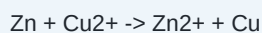
Try to disprove your answer: does each role agree with the species' own state/electron change? Does electron loss equal gain?

GENERALISE

State a method that would still work if Zn/Cu were replaced by unfamiliar species: identity -> state direction -> electron consequence -> role.

RQ-REDOX-03 | Commit before help

GOVERNED REACTION



FROZEN QUESTION

A learner says: 'Zn is the reducing agent because Zn gets reduced.' Diagnose the error and repair the reasoning.

YOUR PLAN

1 What is being asked? 2 Which species/state evidence matters? 3 Which representation will you use? 4 What must remain true? 5 What is your first move? 6 Why is that move legal?

RQ-REDOX-03 | Progressive concept help

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

H1 ORIENT

Name the target in your own words before calculating or labelling anything.

H2 STRUCTURE / H3 REPRESENT

Track same element -> state lane -> electron consequence. Keep role blank until self-change is secure.

H4 PRINCIPLE / H5 FIRST MOVE

Increase = oxidation; decrease = reduction. First move: write the two before/after lanes.

RQ-REDOX-03 | Solution, falsify, generalise

FULL SOLUTION

The claim reverses Zn's self-change. Zn changes 0 -> +2, so Zn is oxidised; because its electron loss reduces Cu^{2+} , Zn is the reducing agent.

FALSIFY

Try to disprove your answer: does each role agree with the species' own state/electron change? Does electron loss equal gain?

GENERALISE

State a method that would still work if Zn/Cu were replaced by unfamiliar species: identity -> state direction -> electron consequence -> role.

Teach-back / transfer reflection

TEACH BACK

Explain the problem-solving strategy without using the words Zn or Cu. Your explanation must tell another learner how to select a first move and how to verify the final role labels.

TARGET STRATEGY

Track identity; choose oxidation-state lanes; infer oxidation/reduction from direction; translate to electron loss/gain; derive agent roles; close the electron ledger; test for a role/self-change reversal.